EVALUATION

ZFOD8

Choose the correct answer:

1. Concentration of the Ag^+ ions in a saturated solution of $Ag_2C_2O_4$ is 2.24×10^{-4} mol L^{-1} solubility product of $Ag_2C_2O_4$ is (NEET – 2017)

   a) 2.42×10^{-8} mol^3L^{-3} b) 2.66×10^{-12} mol^3L^{-3}

   c) 4.5×10^{-11} mol^3L^{-3} d) 5.619×10^{-12} mol^3L^{-3}

2. Following solutions were prepared by mixing different volumes of NaOH of HCl different concentrations. (NEET – 2018)

   i. 60 mL $\frac{M}{10}$ HCl + 40mL $\frac{M}{10}$ NaOH ii. 55 mL $\frac{M}{10}$ HCl + 45 mL $\frac{M}{10}$ NaOH

   iii. 75 mL $\frac{M}{5}$ HCl + 25mL $\frac{M}{5}$ NaOH iv. 100 mL $\frac{M}{10}$ HCl + 100 mL $\frac{M}{10}$ NaOH

   pH of which one of them will be equal to 1?

   a) iv b) i c) ii d) iii

3. The solubility of $BaSO_4$ in water is $2.42\times10^{-3}gL^{-1}$ at 298K. The value of its solubility product (K_{sp}) will be (NEET -2018). (Given molar mass of $BaSO_4$=233g mol^{-1})

   a) $1.08\times10^{-14}mol^2L^{-2}$ b) $1.08\times10^{-12}mol^2L^{-2}$

   c) $1.08\times10^{-10}mol^2L^{-2}$ d) $1.08\times10^{-8}mol^2L^{-2}$

4. pH of a saturated solution of $Ca(OH)_2$ is 9. The Solubility product (K_{sp}) of $Ca(OH)_2$

   a) 0.5×10^{-15} b) 0.25×10^{-10}

   c) 0.125×10^{-15} d) 0.5×10^{-10}

5. Conjugate base for Bronsted acids H_2O and HF are

   a) OH^- and H_2FH^+, respectively b) H_3O^+ and F^-, respectively

   c) OH^- and F^-, respectively d) H_3O^+ and H_2F^+, respectively

6. Which will make basic buffer?

   a) 50 mL of 0.1M NaOH+25mL of 0.1M CH_3COOH

   b) 100 mL of 0.1M CH_3COOH+100 mL of 0.1M NH_4OH

   c) 100 mL of 0.1M HCl+200 mL of 0.1M NH_4OH

   d) 100 mL of 0.1M HCl+100 mL of 0.1M NaOH

XII U8-Ionic equilibrium.indd 28 16-12-2022 19:34:30